

Optimization in Decentralized Learning with Adaptive Neighbor Sampling

Alessandro Bombarda, Matteo Busato, Jacopo Senoner
School of Computer and Communication Sciences, EPFL, Switzerland

Abstract—Decentralized learning avoids a parameter server but requires peer-to-peer model exchange. Recent work shows that uniformly pulling a logarithmic number of neighbors can reduce communication while retaining convergence and robustness guarantees; we ask whether adaptive sampling can improve on this uniform baseline under heterogeneous data. We formulate neighbor selection as a combinatorial semi-bandit problem and compare different no-regret samplers under two reward signals. Our contributions are three empirical observations: (i) adaptive samplers recover latent cluster structure from local feedback; (ii) this recovery induces a personalization–generalization frontier, with aggressive samplers such as CTS favoring local accuracy and milder samplers such as EXP3 or discounted variants preserving more global mixing; (iii) the reward shape affects the algorithm’s position on the frontier; (iv) in static clusters, discounting weakens cluster recovery by forgetting useful historical evidence.

I. INTRODUCTION

In federated learning a server aggregates client updates; a serverless decentralized system removes this coordination point, but information must instead spread through peer-to-peer exchanges, and dense decentralized methods incur message complexity that grows quadratically in the number of nodes. Robust Pull-based Epidemic Learning (RPEL) [3] reduces this burden by having each node pull models from a small, uniformly random subset of peers; its analysis requires only $O(\log n)$ sampled peers and yields $O(n \log n)$ messages per round rather than $O(n^2)$, while preserving convergence and Byzantine-robustness guarantees with high probability.

Uniform sampling is deliberately agnostic to data heterogeneity. When clients form latent groups, however, some peers provide systematically more useful model updates than others, and averaging indiscriminately mixes incompatible objectives and harms per-client performance. This motivates our question: *when does adaptive neighbor sampling improve learning at a fixed communication budget, and when is uniform sampling preferable?* Selecting which neighbors to pull is naturally an online decision: a node does not know in advance which peers are useful and must trade exploration against exploitation. We therefore model every candidate neighbor as an arm; at each round a node selects s arms and observes one reward per selected neighbor, giving a combinatorial semi-bandit.

a) Research questions: (Q1) Does bandit-based neighbor selection improve performance over uniform random sampling and a fixed d -regular topology at equal communication, and how does this depend on heterogeneity? (Q2) Does the converged sampling distribution recover the latent cluster structure

of the partition? (Q3) Which reward signal best drives the bandit, and does decreasing realized regret correspond to better learning? (Q4) How does discounting reposition a sampler on the personalization–generalization front?

b) Contributions: (i) a controlled comparison of adaptive sampler families against uniform sampling and a fixed topology, under an identical communication budget, reporting both accuracy and communication efficiency; (ii) an evaluation across heterogeneous partitions with cluster-level and node-level structure, testing whether the learned distribution recovers cluster structure; (iii) diagnostics that separate downstream learning quality from reward maximization and graph-structure recovery, with the key finding that the three can diverge; (iv) finally, we study the relationship between discounting and the ability of bandit algorithms to recover the latent cluster structure.

II. METHOD

A. Related work

Our problem sits between three lines of work. *Neighbor selection* starts from the observation that, under non-i.i.d. data, not all neighbors are equally useful and uniform gossip is suboptimal: AFIND+ [1] makes sampling adaptive through gradient similarity (using the featurizer output as a proxy for a client’s distribution), adapts the number of sampled neighbors via an entropy-based threshold, and aggregates with contribution-aware weights. *Bandit formulations* make the selection explicit: DFedDC [2] casts dynamic neighbor selection as a multi-armed bandit with a dynamic ε -greedy policy, comparing clients through Dynamic Time Warping of gradient-norm sequences and aggregating by confidence. A *complementary lever* changes how updates are weighted rather than whom to sample: FedAAW [4] derives aggregation weights analytically by minimizing a convergence bound, but in the centralized (server-based) setting. Our starting point is RPEL [3], which samples peers uniformly at random in a pull-based epidemic protocol to obtain robustness at $O(n \log n)$ communication; its sampling is deliberately preference-free, which is exactly the design choice we interrogate.

B. Decentralized learning and feedback

We consider n nodes in a fully-connected network topology. Node i holds local data from distribution $\mathcal{D}_i$ and parameters w_i^t . In round t , every node performs one or more local SGD steps, selects a subset $S_i^t \subseteq [n] \setminus \{i\}$ of size s , pulls those

models, and averages them with its own model. Communication per round is thus of ns model transmissions for every compared sampler; full communication would use $n(n-1)$.

After sampling, node i observes rewards r_j for $j \in S_i^t$. We study two reward models: **inverse model distance** (for short **IMD**) and **cosine similarity** (for short **COS**) defined respectively as:

$$r_{i,j,t}^{\text{dist}} = (1 + \|w_i^t - w_j^t\|_2)^{-1},$$

$$r_{i,j,t}^{\text{upd}} = \frac{1}{2} \left(1 + \frac{\langle \Delta w_i^t, \Delta w_j^t \rangle}{\|\Delta w_i^t\|_2 \|\Delta w_j^t\|_2} \right),$$

where $\Delta w_i^t = w_i^{t-\frac{1}{2}} - w_i^{t-1}$ is the pre-aggregation local update. These rewards encode model compatibility, not direct loss reduction, so their alignment with the learning objective is an empirical question we study in IV.

C. Neighbor samplers

Our baseline is the uniform distribution among s distinct neighbors. We compare ϵ -GREEDY, EXP3, CUCB, and CTS. CUCB selects the s largest upper confidence indices; CTS selects the s largest independent Beta posterior draws. Discounted CUCB and CTS multiply sufficient statistics by a discount $\gamma < 1$ before each update, prioritizing recent observations [19]. In our experiments, we hold $\gamma = 0.99$, and we do separate ablations on this value in appendix D. All methods are compared across identical data splits, initialization protocol, local optimization, and communication budget.

III. EXPERIMENTAL DESIGN

Data heterogeneity. We use pooled FEMNIST with two synthetic partition families. Dirichlet partitions draw class proportions with concentration $\alpha \in \{0.5, 1, 10, 100\}$; smaller α means stronger heterogeneity. Pathological partitions assign k labels to each group, with overlap o between consecutive groups. A group can be a latent cluster whose nodes share its samples uniformly, or a single node. We additionally use FEMNIST’s natural writer identities [20], assigning k writers per node and holding out complete writers for global evaluation. Finally, in Appendix D we report some additional Cifar [21] experiments to verify that our findings hold on a harder optimization landscape.

Controlled factors. The primary study uses $n = 30$ honest nodes, mean aggregation, one local SGD step per round, and sampling fractions $s/(n-1) \in \{0.1, 0.2, 0.4\}$. FEMNIST runs for 1000 rounds, with batch size 64, SGD learning rate 0.05, momentum 0.9, and weight decay 10^{-4} . Validation is performed every 20 rounds. Each configuration is repeated over 3 seeds, and the relevant metrics are averaged over these seeds, except for the clustering plots, for visualization reasons.

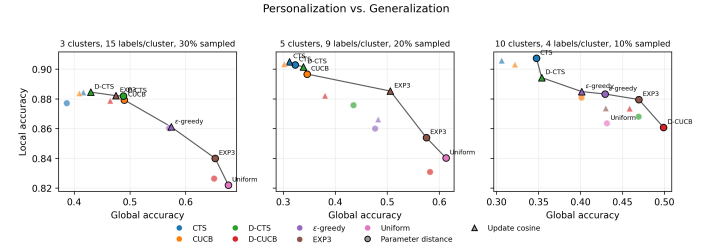
Metrics and fair comparison. The primary endpoint is mean global test accuracy at a fixed number of transmitted models. We also report the worst-node accuracy and the communication required to reach a fixed target accuracy. To explain outcomes, we measure model disagreement, consensus drift, and probability mass assigned to same-cluster peers.

Finally, we report realized reward regret against the best fixed subset in hindsight: $R_i(T) := \max_{|S|=s} \sum_{t=1}^T \bar{r}_{i,S,t} - \sum_{t=1}^T \bar{r}_{i,S_i^t,t}$. This diagnostic compares methods on the trajectory each method actually created.

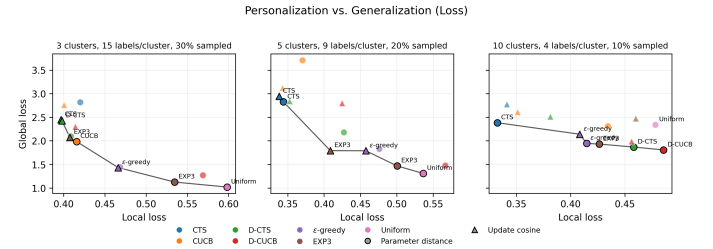
IV. RESULTS AND DISCUSSION

Does adaptation improve communication efficiency?

We report here a Pareto frontier, showing the personalization vs generalization tradeoff between the various samplers. With less heterogeneity (3-5 clusters), the uniform is Pareto-optimal, while it stops being so with 10 clusters. We investigate the whole optimization process over the rounds, and we report these detailed plots in Appendix C.2.



(a) Local validation vs global test accuracy.



(b) Local validation vs global test loss.

Fig. 1: Pareto frontier, global accuracy vs local, of the chosen samplers. Sampling ratio is adapted to cluster size (sampled = cluster size + 1)

Do bandits recover useful latent structure? Figure 2 shows the adaptive samplers are in general capable of recovering the latent cluster structures. Comparing with Figure 1 we can see that the preference intensity is correlated with the personalization-generalization tradeoff: CTS aggressively samples cluster neighbors and has strong personalization but poor generalization, whereas EXP3 generalizes better, but loses on personalization. The uniform baseline has no preference.

How well does reward proxy the optimization target? The reward signal drives the bandit algorithm’s sampling choices, so it is natural to ask whether the rewards align with the optimization objective.

Table I shows a clear split: COS is a better proxy for personalization, whereas IMD better preserves generalization. Switching from IMD to COS improves local accuracy in 16 of 18 pathological configurations, but degrades global accuracy in 15 of 18. The loss metrics tell the same story: COS lowers local loss in 16 of 18 cases, but lowers global loss in only 4. The reward comparison histograms in Fig. 4 show the

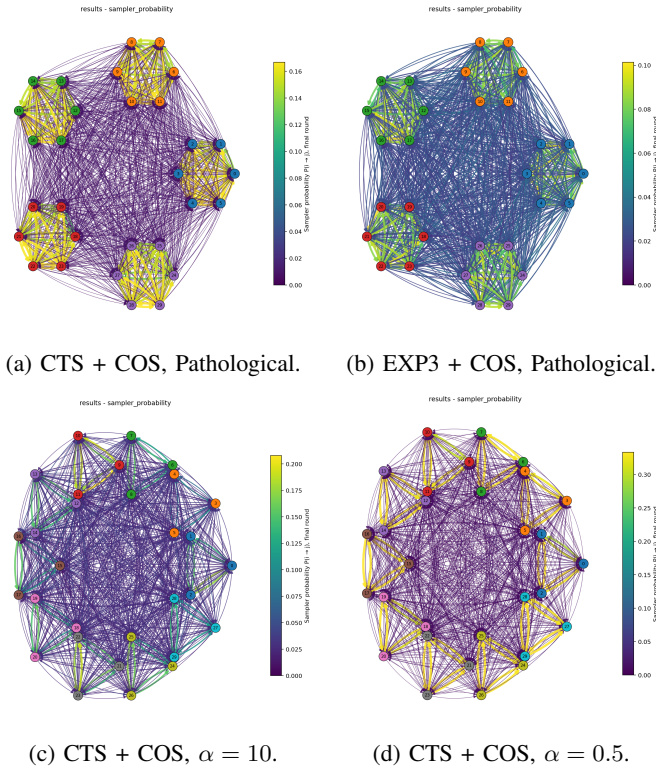


Fig. 2: Final sampling probabilities for pathological- and Dirichlet-based FEMNIST clusters.

magnitude: COS usually moves the operating point toward higher local accuracy and lower local loss, at the cost of lower global accuracy and higher global loss.

This behavior is expected: COS scores the direction of a node’s update, so it better reflects local task similarity and encourages same-cluster sampling. IMD instead favors nearby models, which keeps more cross-cluster mixing and therefore better preserves global accuracy. Figure 6 illustrates this mechanism for EXP3. The alpha-cluster sweep confirms that the COS advantage weakens as heterogeneity decreases. At $\alpha = 0.5$, COS improves local accuracy in 16 of 18 paired configurations and improves global accuracy in only 5 of 18. At $\alpha = 10$, the effect essentially disappears: local accuracy improves in 10 of 18 cases, global accuracy in 8 of 18, and the mean changes are below 0.1 percentage points. Thus COS is better than IMD as a proxy for personalization when a stable cluster structure exists; it is not a universal proxy for global optimization. Appendix C.1 gives the full breakdown.

When does discounting fail?

While the reward decides which peers look useful, the discount factor decides how long evidence about those peers is remembered. In our static-cluster setting, useful peers are persistent: a node’s cluster identity does not change during training. Undiscounted CTS and CUCB can therefore accumulate evidence about same-cluster neighbors, while discounted variants down-weight that evidence and relax the sampling distribution toward broader mixing.

TABLE I: How often COS improves over IMD in paired adaptive configurations on pathological FEMNIST. Results are averaged over network configurations (clusters + sampling + label distribution)

Split	Metric	Improves	Worsens	Mean Δ
Pathological	Local acc.	16/18	2/18	+1.60 pts
Pathological	Global acc.	3/18	15/18	−6.84 pts
Pathological	Local loss	16/18	2/18	−0.047
Pathological	Global loss	4/18	14/18	+0.376
$\alpha = 0.5$	Local acc.	16/18	2/18	+0.72 pts
$\alpha = 0.5$	Global acc.	5/18	13/18	−1.32 pts
$\alpha = 10$	Local acc.	10/18	8/18	+0.03 pts
$\alpha = 10$	Global acc.	8/18	10/18	−0.06 pts

The results supporting these claims are in Table II, with a CIFAR ablation. Under IMD, decreasing γ from 1 to 0.9 lowers trajectory-averaged local accuracy from 58.1% to 33.7% for CTS and from 44.4% to 26.9% for CUCB, while moving the sampling graph closer to uniform (Fig. 9, see Appendix D).

TABLE II: Discounted vs. non-discounted samplers under the IMD reward. 30 nodes, CIFAR-10, pathological 5g/2c heterogeneity, 500 rounds, seed 0. Values are trajectory averages over rounds and nodes (same reduction as the heatmaps). $^\dagger \gamma=1.0$ is non-discounted. Best per column in **bold**.

Sampler	γ	Val. Acc $\uparrow$	Global Acc $\uparrow$	Val. Loss $\downarrow$	Regret $\downarrow$
CUCB	1.0 †	0.444	0.111	1.312	24.7
	0.99	0.367	0.119	1.685	31.2
	0.95	0.253	0.130	2.065	37.5
	0.9	0.269	0.155	1.986	37.4
CTS	1.0 †	0.581	0.132	1.013	15.9
	0.99	0.550	0.127	1.073	18.9
	0.95	0.393	0.119	1.507	30.2
	0.9	0.337	0.126	1.773	35.7

V. CONCLUSION

We formulated adaptive neighbor selection in decentralized learning as a combinatorial semi-bandit problem. We’ve designed two different rewards and shown that, under clustered heterogeneity, no-regret samplers can recover latent structure, and lie on a personalization–generalization frontier: aggressive ones such as CTS favor local specialization, while milder ones such as EXP3 preserve broader mixing and global performance. We also found that reward design affects the operating mode, with COS favoring personalization and IMD being more aligned with generalization. Finally, we found discounting to be detrimental to personalization, when useful relationships remain stationary.

Future work should test whether samplers discover useful relationships under node-level heterogeneity, where no cluster structure is imposed. It should also evaluate robustness against Byzantine participants, measuring whether adaptive samplers learn to avoid malicious updates and how this affects the communication–accuracy trade-off.

REFERENCES

- [1] Smart Sampling: Helping from Friendly Neighbors for Decentralized Federated Learning. <https://arxiv.org/pdf/2407.04460>.
- [2] DFL with Dynamic Neighbor Selection and Confidence-Based Aggregation (DFedDC). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5560620.
- [3] Robust and Efficient Collaborative Learning (RPEL). <https://arxiv.org/pdf/2510.08311>.
- [4] FL with Adaptive Aggregation Weights for Non-IID Data in Edge Networks (FedAAW). <https://ieeexplore.ieee.org/document/10854512>.
- [5] Richard H. Kallet, "How to Write the Methods Section of a Research Paper," *Respiratory Care*, vol. 49, no. 10, pp. 1229–1232, 2004.
- [6] H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," arXiv:1602.05629, 2023. <https://arxiv.org/abs/1602.05629>.
- [7] Greg Anderson, "How to Write a Paper in Scientific Journal Style and Format," Bates College, 2004. <http://abacus.bates.edu/~ganderso/biology/resources/writing/HTWtoc.html>.
- [8] Simon Peyton Jones, "How to Write a Great Research Paper," Microsoft Research Cambridge, 2008.
- [9] Editorial, "Scientific Writing 101," *Nature Structural Molecular Biology*, vol. 17, p. 139, 2010.
- [10] Jonathan B. Buckheit and David L. Donoho, "WaveLab and Reproducible Research," Stanford University, Tech. Rep., 2009.
- [11] Robert Gentleman, "Reproducible Research: A Bioinformatics Case Study," *Statistical Applications in Genetics and Molecular Biology*, vol. 4, no. 1, 2005. <http://www.bepress.com/sagmb/vol4/iss1/art2>.
- [12] Matthias Schwab, Martin Karrenbach, and Jon Claerbout, "Making Scientific Computations Reproducible," *Computing in Science Engineering*, vol. 2, no. 6, pp. 61–67, 2000. <https://doi.org/10.1109/5992.881708>.
- [13] Joel Spolsky, *Joel on Software: And on Diverse Occasionally Related Matters That Will Prove of Interest etc.: And on Diverse and Occasionally Related Matters ... or Ill-Luck, Work with Them in Some Capacity*. APRESS, 2004.
- [14] Andrew Hunt and David Thomas, *The Pragmatic Programmer*. Addison Wesley, 1999.
- [15] Alex Krizhevsky, "Learning Multiple Layers of Features from Tiny Images," University of Toronto, Tech. Rep., 2009. <https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf>.
- [16] P. Auer, N. Cesa-Bianchi, Y. Freund, and R. E. Schapire, "The non-stochastic multiarmed bandit problem," *SIAM Journal on Computing*, vol. 32, no. 1, pp. 48–77, 2002.
- [17] W. Chen, Y. Wang, and Y. Yuan, "Combinatorial multi-armed bandit: General framework and applications," in *International Conference on Machine Learning (ICML)*, 2013, pp. 151–159.
- [18] A. Huyuk and C. Tekin, "Analysis of combinatorial Thompson sampling for combinatorial multi-armed bandit problems," *IEEE Transactions on Signal Processing*, vol. 67, no. 4, pp. 933–948, 2018.
- [19] A. Garivier and E. Moulines, "On upper-confidence bound policies for non-stationary bandit problems," *arXiv preprint arXiv:0805.3415*, 2011.
- [20] S. Caldas, S. M. K. P. Wu, T. Li, J. Konečný, H. B. McMahan, V. Smith, and A. Talwalkar, "LEAF: A benchmark for federated settings," *arXiv preprint arXiv:1812.01097*, 2018.
- [21] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009.

APPENDIX A IMPLEMENTATION DETAILS

A.1 Model details

All convolutional networks use ReLU activations, max-pooling, and SGD training. See Table III. For CIFAR-10, convolutions use same padding and max-pooling is applied after each block.

A.2 Sampling details

All experiments use $m = 30$ clients. Each client samples $k = \lfloor 0.2 \times 30 \rfloor = 6$ neighbors per round (20% sampling budget), unless otherwise specified. Local steps per round: $\tau = 1$ (a single local SGD step before aggregation).

A.3 Optimization details, communication rounds

For FEMNIST: SGD with momentum 0.9, learning rate 0.05, weight decay 10^{-4} , batch size 25, NLL loss, 1000 communication rounds. For CIFAR-10: SGD with momentum 0.99, learning rate 0.5, weight decay 10^{-2} , batch size 50, CrossEntropyLoss, 500 communication rounds.

A.4 Evaluation details, metrics

The global test set is a 10% holdout of the full training pool (before client partitioning); each client additionally holds out 20% of its own partition as a local test set. Metrics are evaluated every 20 rounds.

TABLE III: Neural network architectures used for each dataset.

Dataset	Architecture	Main layers	Parameters
MNIST	2-conv / 2-FC	20/50 conv filters, 5x5 kernels; dense layers 500/10	$\approx 431\text{K}$
FEMNIST	2-conv / 2-FC	64/128 conv filters, 5x5 kernels; dense layers 1024/62	$\approx 2.37\text{M}$
CIFAR-10	3-conv / 3-FC	20/100/200 conv filters, 5x5 kernels; dense layers 512/256/10	$\approx 2.32\text{M}$

APPENDIX B PARTITIONING

B.1 Data heterogeneity: partition scheme

Let C be the number of classes and G the number of groups ($G \leq m$, must divide m). For each class $c \in \{1, \dots, C\}$ draw independently

$$\mathbf{p}_c \sim \text{Dir}(\underbrace{\alpha, \dots, \alpha}_G), \quad \mathbf{p}_c \in \Delta^{G-1}, \quad (1)$$

and assemble the $G \times C$ heterogeneity matrix M with $M_{gc} = p_c[g]$, so each column sums to one. Smaller α concentrates mass on fewer groups per class (high heterogeneity); $\alpha \rightarrow \infty$ recovers uniform i.i.d. allocation.

B.2 Pathological partition.

As an alternative to Eq. (1), a sparse binary mask assigns exactly L consecutive label indices to each group, with $s = L - \text{overlap}$:

$$M_{gc} = \mathbf{1}[c \in \{(g \cdot s + j) \bmod C : j = 0, \dots, L - 1\}] \quad (2)$$

after column-normalizing so each label is fully consumed. This yields hard, near-disjoint clusters that share exactly *overlap* labels between adjacent groups.

B.3 Hierarchical assignment.

Algorithm 1 shows how M is turned into per-worker index sets. Workers within the same group receive an i.i.d. subsample of their group’s pool, so the two-level structure lets $G < m$ impose cluster structure while keeping within-cluster variance low.

APPENDIX C FURTHER RESULTS ON FEMNIST

C.1 Rewards and samplers

In Figure 3 we present the fully detailed heatmap for global and local accuracy on our pathological-clusters sweep on FEMNIST. Figure 4 shows the performance of the COS vs IMD rewards, averaged over samplers both on dirichlet-clusters and pathological-clusters. We can see that a higher heterogeneity favours COS rewards for personalization, whereas at lower heterogeneity the difference is not significant.

Algorithm 1 Hierarchical data partition (used by both methods)

Require: targets $\mathbf{y}$, matrix $M \in \mathbb{R}^{G \times C}$, workers per group $w = m/G$

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1: for each class  $c$  do
2:    $I_c \leftarrow$  shuffled indices  $\{i : y_i = c\}$ 
3:   for each group  $g$  do
4:      $\text{count}_{gc} \leftarrow \lfloor M_{gc} \cdot |I_c| \rfloor$ 
5:   end for
6:   distribute  $I_c$  into group pools  $\mathcal{P}_g$  according to  $\text{count}_{gc}$ 
7: end for
8: for each group  $g$  do
9:   shuffle  $\mathcal{P}_g$ ; split into  $w$  equal chunks  $\rightarrow$  workers  $\{i : \lfloor i/w \rfloor = g\}$ 
10: end for

```



Fig. 3: First row of heatmaps: local accuracy; second row: Global accuracy. For the heatmap, rows represent rewards, while columns are samplers. Detailed observations are presented in Section IV. Uniform is reported for completeness under IMD, but it's reward-independent.

C.2 Detailed results on training process

We now present detailed results on the training process, focusing on how the studied metrics evolve across rounds.

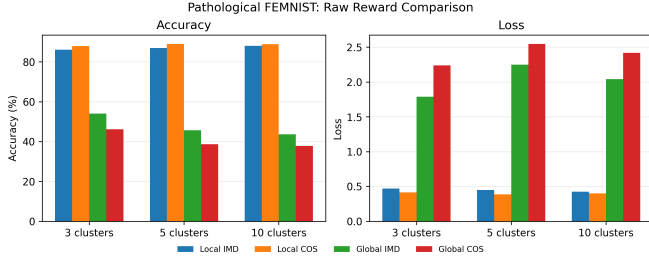
As expected, the bandit algorithms that achieve the highest local accuracy are also those that better prevent drift among nodes. This results in smaller differences across the considered metrics, including regret, local accuracy, and neighbor disagreement.

APPENDIX D FURTHER RESULTS ON CIFAR

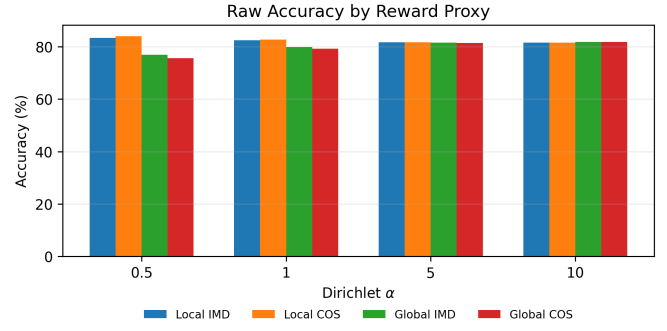
D.1 Cifar experiments: sweeping on bandit algorithms and discounts

We provide additional experiments on the Cifar dataset, whose goal is to further test our Bandit algorithms and check the performance of discounting.

CUCB with cosine similarity. First, we consider the CUCB bandit with COS reward. The results show that the gradient norm increases over time, while the global test accuracy remains very low and nearly constant. The neighbour disagreement is highly noisy, and the regret does not converge. The time-normalized reward decreases, suggesting that the model weights are drifting apart rather than reaching consensus. The discounted and non-discounted variants behave almost identically, with the exception of neighbour disagreement, which is substantially worse in the discounted case. For this reason, we do not report the discounted results, since they are largely redundant; an explanation for this behaviour is provided below.

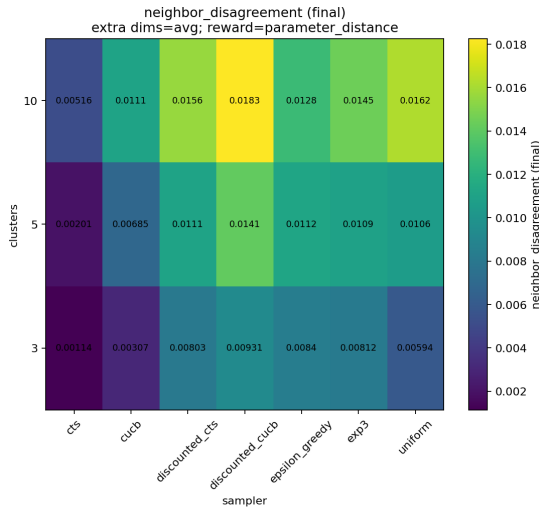


(a) Pathological clusters.

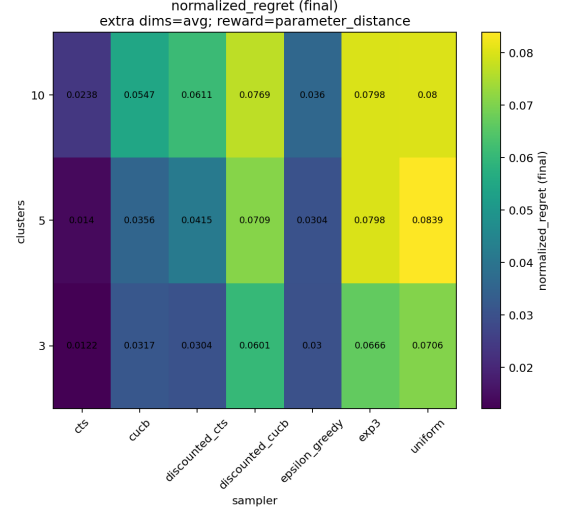


(b) Dirichlet clusters.

Fig. 4: Effect of changing the reward from IMD to COS. Positive accuracy values favor COS; negative loss values are better. COS improves local performance when heterogeneity is strong, but sacrifices global performance.



(a) Neighbour disagreement, IMD reward, FEMNIST.



(b) Regret, IMD reward, FEMNIST.

Fig. 5: FEMNIST, IMD reward. We expect neighbour disagreement to be higher for samplers which are more similar to uniform, and we found this result here, e.g. see that CTS has lower neighbour disagreement than CUCB. Similarly for regret.

CTS and COS the gradient norm initially increases. However, unlike in the CUCB case, the training loss decreases and the local accuracy improves, providing the first signs of convergence. The clustering structure also appears to be detected, although the global test accuracy remains poor. At the same time, both regret and neighbour disagreement continue to increase, indicating that the system is still far from full convergence.

CUCB and CTS with IMD. The detected clusters are much clearer than with the cosine similarity reward. The normalized regret decreases for the first time, see Figure 8 (f), and the gradient norm eventually starts to decrease as well. These results suggest that cosine similarity performs worse than model distance as a reward, as this holds also for CTS (but we will not focus on it). Note that, as it chooses to sample the top K nodes according to its internal weights, and we chose a sample probability of 20%, which is 6 nodes in our setting, the best result would be to sample each node in its own cluster plus another node. This indeed happens.

Note that, although CTS samples much more from its neighbours, its global validation accuracy is equal to the one of CUCB Figure 10, confirming it achieves the best personalization vs generalization results. This is coherent also with the fact that in Figure 10 the inverse of model distance of CTS between nodes not in the same cluster is not much higher than CUCB.

Discount factor. As observed before, increasing the discount factor makes the resulting distribution increasingly similar to the uniform distribution. We therefore report a comparison of the clustering behaviour as the discount is reduced.

CUCB vs CTS. Finally, we compare the clustering capabilities of CUCB and CTS in Figure 10. CUCB performs worse, while CTS shows a moderate improvement. We compare them reward-wise, since CUCB is a deterministic algorithm and thus interpreting its sampled neighbors as probabilities doesn't make much sense.

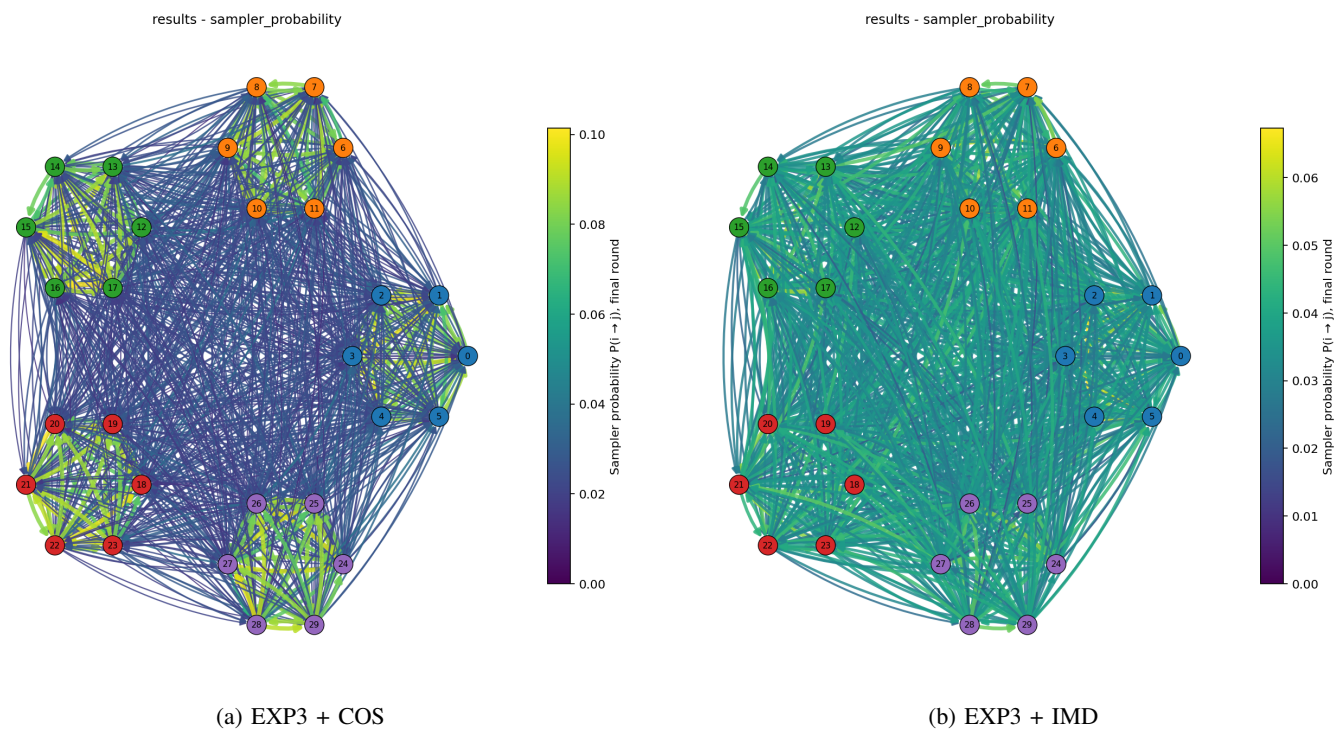
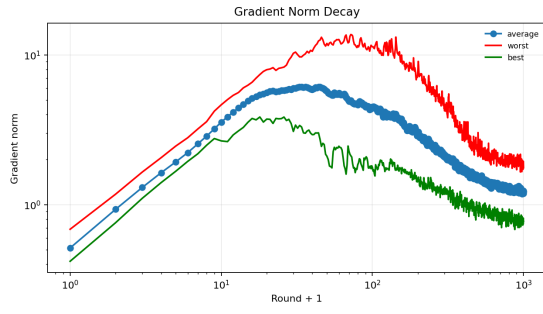
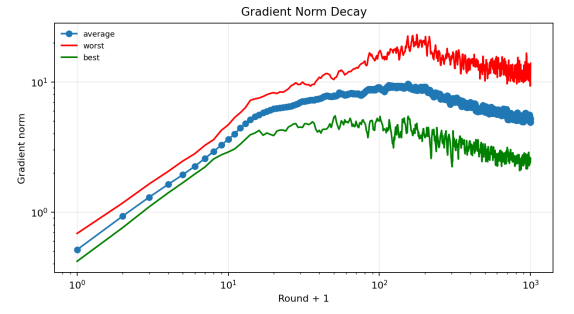


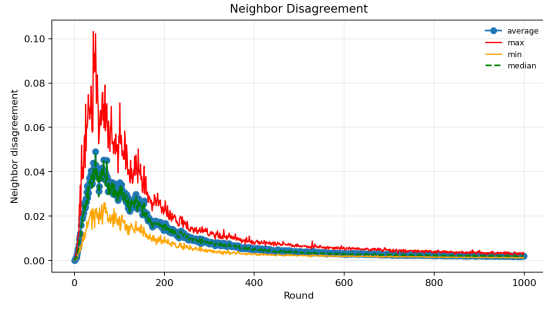
Fig. 6: EXP3 sampling on the pathological-cluster ($c=5$) FEMNIST run with IMD vs COS reward. We can see that COS drives the algorithm to be more aggressive towards sampling in-cluster neighbors.



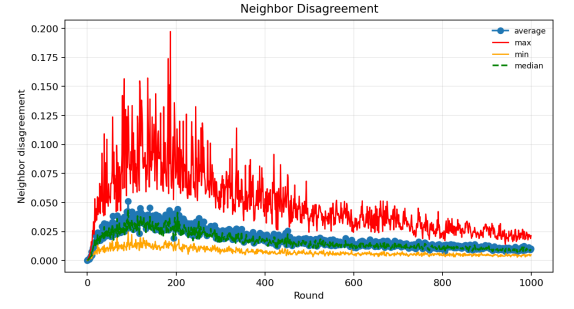
(a) CTS: Gradient norm



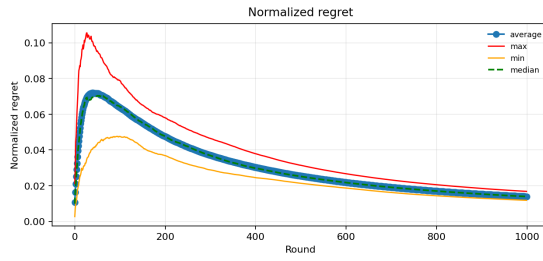
(b) EXP3: Gradient norm



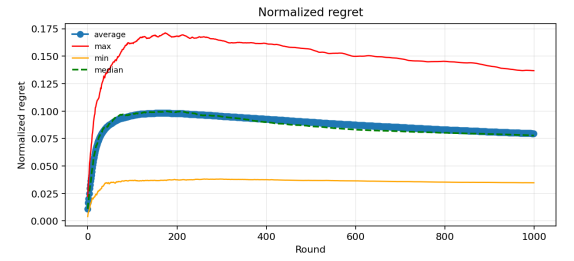
(c) CTS: Neighbor disagreement



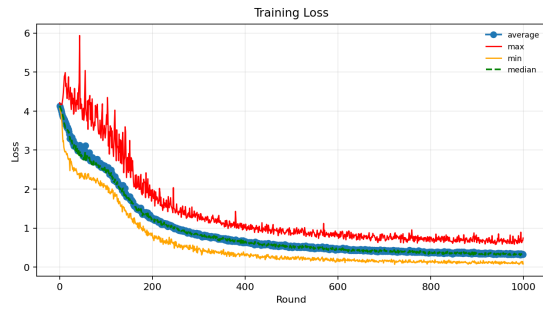
(d) EXP3: Neighbor disagreement



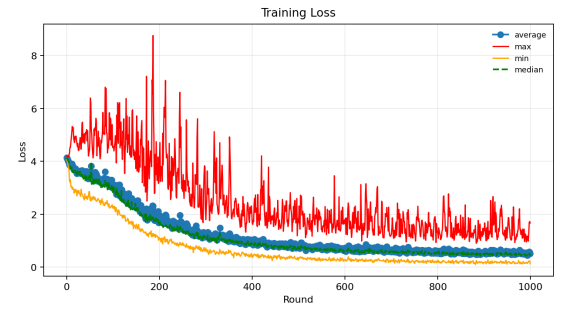
(e) CTS: Regret



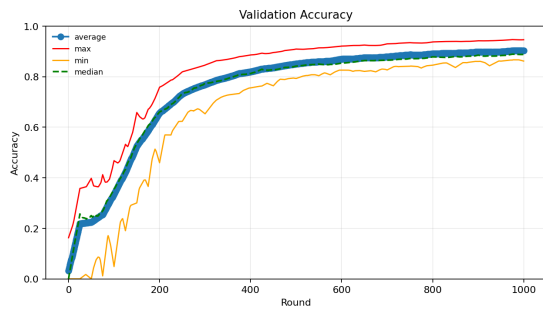
(f) EXP3: Regret



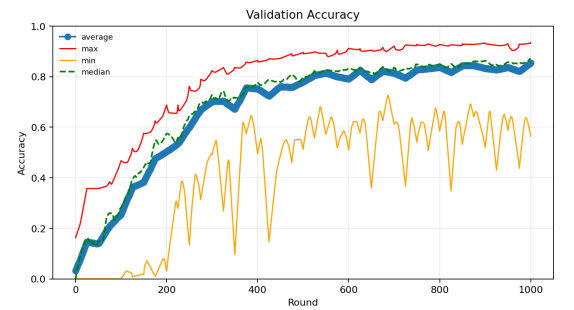
(g) CTS: Train loss



(h) EXP3: Train loss



(i) CTS: Validation accuracy



(j) EXP3: Validation accuracy

Fig. 7: Comparison between CTS and EXP3 metrics on FEMNIST pathological clusters.

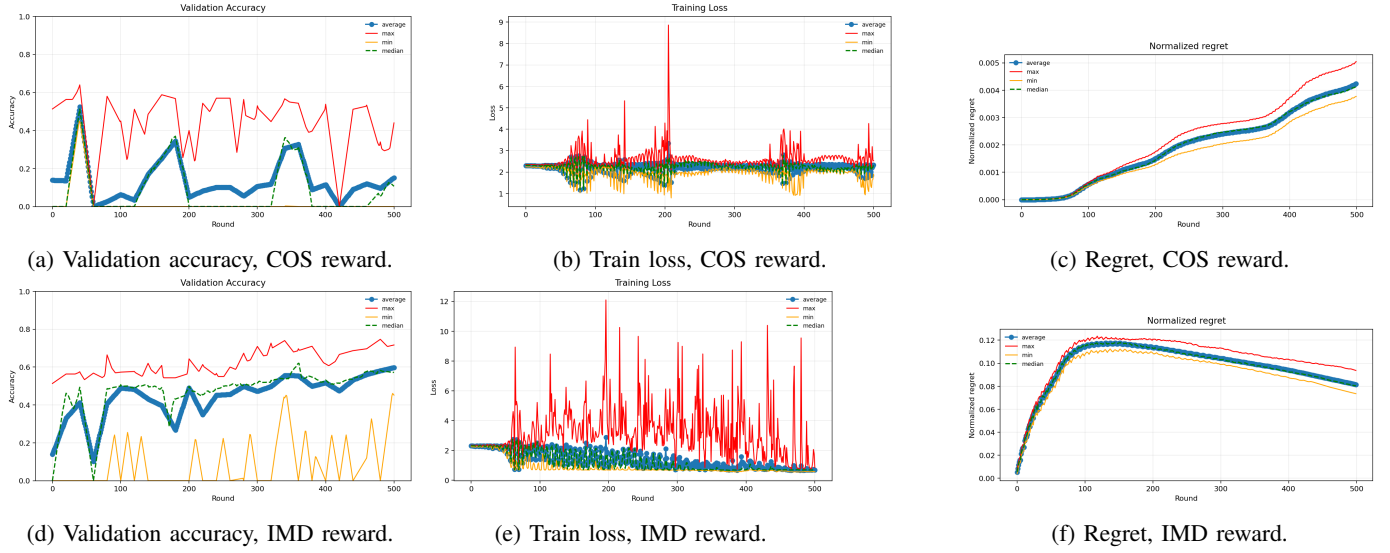


Fig. 8: Performance metrics against rounds for CUCB, cosine similarity reward on first row and model distance on second row, CIFAR. These results suggest cosine similarity performs worse than model distance as a reward.

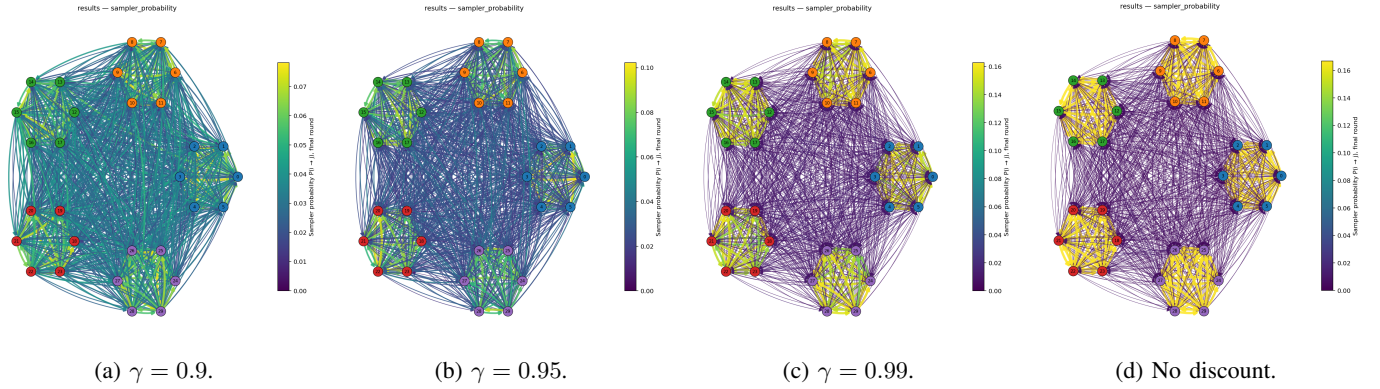


Fig. 9: Comparison of CTS probabilities for different discount factors, final round, **IMD** reward. The result of discounting is that the final distribution is closer to a uniform, and thus less accurately identifies the clusters.

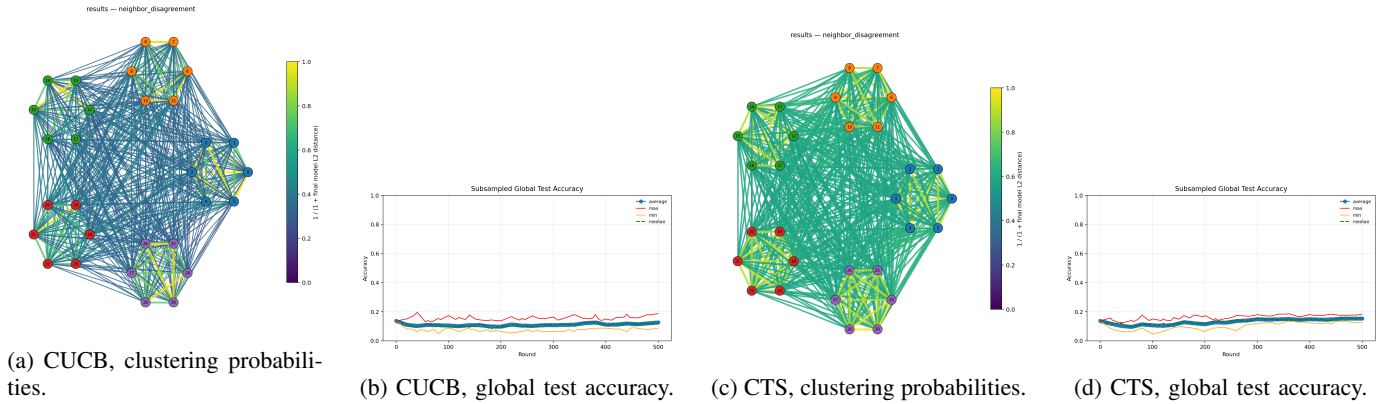


Fig. 10: Comparison CUCB vs CTS with **IMD**-based rewards, final round.